MINGZHE LI

Financial Economics, Household Finance, Continuous Time
Shihezi University ⋄ mingzheli@stu.shzu.edu.cn

WORKING PAPERS

On Apparent Relative Risk Aversion (2026) *Available at* SSRN 6911380

Abstract: For HARA utility satisfying the precautionary saving motive, apparent relative risk aversion—the value function's relative risk aversion, which governs the optimal risky portfolio share—lacks a general characterization under concave consumption. Given a constant $1/(k-1)$, it is characterized by the elasticity of supernumerary consumption with respect to wealth, bounded above by unity. Linear consumption implies that apparent relative risk aversion is increasing and concave, such that the optimal risky portfolio share is decreasing and convex in the Merton model, a pattern difficult to reconcile with limited participation and underdiversification. This paper derives necessary and sufficient conditions for the monotonicity and concavity of apparent relative risk aversion under concave consumption. These conditions provide a benchmark to assess whether deviations rationalize heterogeneous investor behavior.

WORK IN PROGRESS

On Apparent Relative Risk Aversion: Numerical Evidence from Continuous-Time Portfolio Choice (2026)

Abstract: This paper develops a numerical framework to analyze apparent relative risk aversion (ARRA) in a continuous-time portfolio choice problem where closed-form consumption policy will be unavailable. The study investigates how the curvature of the optimal consumption function across the wealth distribution shapes the ARRA profile and the optimal risky portfolio share. Through numerical experiments, this paper evaluates the theoretical benchmarks established in the companion paper, which derives analytical conditions for the monotonicity and concavity of ARRA given a concave optimal consumption rule.

Household Problem with Risk Heterogeneity: A Continuous-Time Framework (2026)

Abstract: The classical framework assumes that the evolution of household welfare is deterministic and does not respond to risk sources, thus failing to distinguish the impact of aggregate risk versus idiosyncratic risk on household financial decisions. This paper relaxes this assumption, allowing the cost functional to have a stochastic response to any risk source. Using the Malliavin derivative as a sensitivity measure, the generalized HJB equation is derived through the Itô-Wentzell-Kunita formula. Compared to the classical framework, the generalized HJB equation includes a new term characterizing the cross-effect between household endowment risk exposure and welfare sensitivity to risk sources. We identify when the classical framework holds: risk sources do not directly affect household endowments (stochastic discounting, preference shocks, bequest functions, etc.), or households have no risk exposure (the Huggett model in continuous time).

COMPUTATIONAL WORK

Julia Implementation of "Financial Frictions and the Wealth Distribution" (2024)

Abstract: Independent implementation and replication of the model in "Financial Frictions and the Wealth Distribution, Fernández-Villaverde et al. (2023)" using Julia.

Open-Source Contribution for Modernization of "Recursive Macroeconomic Theory" Codebase (2024) *Available at* GitHub Repository

Abstract: This repository contains Matlab solutions and updated functions from Ljungqvist & Sargent's textbook (2018, MIT) for compatibility with Matlab 2024+.

EDUCATION

Shihezi University 2024-Present
Master of Finance (M.F.)

Sichuan University 2023
B.Eng. in Computer Science

COMPUTATIONAL METHODS

Julia: Neural Networks, Numerical Analysis & PDEs, Tensor Calculation, Finite Difference Method

Matlab: DSGE

LANGUAGES

Chinese, English

Last updated: July 14, 2026